Boundary Graph Deformations for the Spacetime Penrose Inequality: Exact Ram Differentials, Reflected Flux Measures, and Fixed-Truncation Degree

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Abstract

We study the distinct September 27, 2026 boundary-graph preprint, with emphasis on its three-dimensional existence proof, its four-dimensional regularity mechanism, and its retained-tensor maximal construction. The graph height and transformed conformal variable are coupled: eliminating the scalar equation does not freeze its solution or remove the nonlinear boundary flux. We formulate the exact differential and its reduced Schur complement, prove an interface product identity, and extend a local flux-potential estimate from bounded surface flux to surface L^p flux for $p > n-1$. A critical-exponent example explains the obstruction to the same estimate at $p = n-1$. We also prove a conditional polynomial graph-tail estimate when maximality is relaxed. These are restricted local results, not a replacement for the source's admissible a priori bounds, degree, exhaustion, preparation, or geometric comparison arguments. Literal source-read intervals and finite checks are recorded separately; reading and successful algebra tests do not certify a global proof. Contribution assessment: C2/5, source-specific restricted proofs and exact reformulation with unverified global dependencies.

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1 Scope, Attribution, and Geometric Domains

This report concerns *Boundary graph deformations for the spacetime Penrose inequality*, pinned to the local source commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. It is not the later composite spacetime Penrose source treated in TR-M512. The local build contains twenty TeX files and a bibliography. The complete selected proofs read here are the three-dimensional degree chapter, the four-dimensional regularity chapter, the transition lemma, and the maximal chapter's local regularity proposition. The three-dimensional setting and system and selected maximal statements were also read. Exact intervals, exclusions, and byte identities are in the accompanying read-scope receipt. Unread chapters are dependencies, not verified premises.

We use four attribution levels. An *inherited statement* is reported from the source. An *own restricted result* is proved here with its hypotheses displayed. A *conditional continuum statement* requires a function-space estimate or invertibility not supplied by finite samples. A *finite check* concerns only the exact algebra or sampled model described in its script. Applicability level L4 concerns relevance of the Ram method; it is not the contribution score C2.

1.1 Three-dimensional prepared and original data

The prepared exterior has one asymptotically flat end, compact nonempty boundary, and a smooth metric complete up to that boundary. Full cuts enclose the entire inner boundary and separate it from the selected end. Disconnected cuts, coincident boundary portions, and filling bounded complementary pockets matter; testing only connected or symmetric surfaces changes the problem. Write A_* for the full-cut infimum. The source uses

$$16\pi\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad 8\pi J = \operatorname{div}_g(K - (\operatorname{tr}_g K)g).$$

Prepared data have compactly supported K, all prescribed fixed-order end bounds for $g-\delta = O(r^{-1})$, scalar decay $O(r^{-3-\delta_1})$, a strict weighted dominant-energy margin, positive energy with zero momentum, strict inner trappedness, and $A_* > 0$. The claimed deformation has $R_{\widehat{g}} \geq 0$, strictly negative inner mean curvature, $\widehat{g} \geq e^{-4\varepsilon} g$, and energy error bounded by

$$C(1+N)^A (e^{-\varepsilon N} + e^{-\varepsilon N/2}).$$

These are inherited global outputs, not consequences of the local propositions below.

For the original weakly trapped data the source assumes, in particular, $E > |P|$, suitable integrable constraints, and decay rate $q > 1/2$. Rest-frame preparation and a separate Riemannian comparison are needed to infer $\sqrt{E^2 - |P|^2} \geq \sqrt{A_*/(16\pi)}$. This report does not remove the strict spacelike-energy assumption, classify equality in original data, or prove the preparation theorem.

1.2 Dimension four and the retained-tensor branch

In dimension four the source normalizes $\mu = (R_g + (\operatorname{tr} K)^2 - |K|^2)/2$ and $J = \operatorname{div}_g(K - (\operatorname{tr} K) g)$. The energy and momentum normalization denominators are respectively $6\omega_3$ and $3\omega_3$, with $\omega_3 = 2\pi^2$. The prepared compact-tensor branch has $g - \delta = O(r^{-2})$ and scalar decay $O(r^{-4-\delta_1})$ with all required fixed-order derivative bounds. Here $\widehat{g} \geq e^{-2\varepsilon} g$ implies that every three-dimensional cut has volume at least $e^{-3\varepsilon}$ times its original volume. The $2/3$ power in the mass-area comparison produces an $e^{-2\varepsilon}$ loss. This is an all-cut comparison; it does not establish existence or regularity of the minimizing enclosure.

The maximal vacuum application has a ball-exterior topology, connected S^3 boundary, an effective T^2 action of the specified asymptotic type, a marginally outer trapped boundary, outermost and outer-area-minimizing hypotheses, $P = 0$, and $E > 0$. The source distinguishes these application hypotheses from those used in the construction itself. Outer-area-minimizing means among all admissible cuts, including noninvariant and disconnected ones. The retained-tensor construction uses

$$1 < q < 2, \quad 0 < \delta < q - 1, \quad \operatorname{tr}_g K = 0, \quad g - \delta = O_2(r^{-q}), \quad K = O_1(r^{-1-q}),$$

with strict prepared inequalities and separate all-fixed-order bounds on unit end patches. Smoothness plus the displayed two/one derivative bounds does not imply those uniform higher-order estimates. Keeping a decaying K is not the same branch as replacing K by compact support.

2 Exact Variables and the Coupled Operator

Fix the background metric and tensor, a finite truncation Ω_R, a deformation parameter N, and one homotopy parameter. Put $c = 4$ in dimension three and $c = 3$ in dimension four. All derivatives below keep this background and these parameters fixed. To avoid confusing the gradient with the homotopy parameter, set $a = \nabla f$.

$$p(t) = N\vartheta(Nt), \qquad l(t) = \exp \int_0^t p(z)\, dz, \qquad L(t) = e^{2t} l(t), \tag{1}$$

$$D = 1 + l^2 |a|^2, \qquad d = D^{-1}, \qquad w = \frac{la}{\sqrt{D}}, \quad v = |w|^2 = 1 - d, \tag{2}$$

$$Z = t + \frac{\log D}{2c}, \qquad \chi = \frac{cd}{c + pv}, \qquad b = \frac{c+p}{c+pv}, \tag{3}$$

$$A_\chi = I - bw \otimes w, \qquad A_d = I - w \otimes w, \qquad u = L\sqrt{D}. \tag{4}$$

The transverse eigenvalues of A_χ are one and its axial eigenvalue is χ: indeed $1 - bv = \chi$. In the source's nonnegative-p regime the map $t \mapsto Z$ at fixed a satisfies

$$\partial_t Z = 1 + pv/c > 0. \tag{5}$$

Its inversion therefore precedes, and does not assume, the admissibility floor $t > -\varepsilon$. Let

$$H_f = \frac{l}{\sqrt{D}}\nabla^2 f, \qquad S = K + H_f, \qquad \mathcal{F} = A_\chi : S.$$

Write $\mathcal{P}$ for the prescribed scalar right-hand side. At an actual solution $\mathcal{F} = \mathcal{P}$; off a solution they are different functions. In the first homotopy stages $\mathcal{P}$ contains the height and collar penalties, while in the final scalar stage it contains the axial trace of K and a directional penalty. The scalar residual is $\mathcal{R}_1 = \mathcal{F} - \mathcal{P}$ with its prescribed Dirichlet data.

For a fixed stage parameter λ, the flux is

$$V_\lambda = uA_\chi(c\nabla Z + \lambda K w), \qquad \operatorname{div}_g V_\lambda = u\Xi. \tag{6}$$

The normal ν on the inner boundary points *into* the exterior; it is the negative of the outward normal of Ω_R there. If $V_\lambda \cdot \nu = q$, the weak residual, for tests vanishing on the outer boundary, is

$$\mathcal{R}_2(f, Z)[\psi] = \int_{\Omega_R} V_\lambda \cdot \nabla\psi + \int_{\Omega_R} u\Xi\psi + \int_{\partial_-\Omega_R} q\psi. \tag{7}$$

The plus sign of the last integral is essential. The source arranges q using the *prescribed* $\mathcal{P}$, not the second-derivative expression $\mathcal{F}$, so that the normal datum depends on $f, Z, \nabla f$ but not $\nabla^2 f$. That choice is used in boundary regularity and cannot be discarded in differentiation.

2.1 The actual source term, not a coercive substitute

For dimension three let $e = a/|a|$ when $a \neq 0$, set $\omega = |w|$, and decompose S as (T, M, j_*) on $e^\perp \oplus \mathbb{R}e$, with $\operatorname{tr} T = \mathcal{F} - \chi j_*$. Put $x = \partial_e t$, $y = \nabla_\perp t$. The source's exact quadratic expression is

$$\begin{aligned}\mathcal{T} = {} & \tfrac{1}{2}|T^{\mathrm{tf}}|^2 + (M + p\omega y)\cdot(M + (p+2)\omega y) + 4(p+1)(|y|^2 + dx^2) \\ & + (\chi - \chi^2/4)j_*^2 + 2(p+1)\chi j_*\omega x - \chi\mathcal{F}j_*/2 + 3\mathcal{F}^2/4.\end{aligned} \tag{8}$$

Completing the squares gives the inherited, directly expandable identity

$$\begin{aligned}\mathcal{T} = {} & \tfrac{1}{2}|T^{\mathrm{tf}}|^2 + |M + (p+1)\omega y|^2 + [4(p+1) - v]|y|^2 \\ & + 4(p+1)d\left(x + \frac{\chi\omega j_*}{4d}\right)^2 + \tfrac{3}{4}(\mathcal{F} - \chi j_*/3)^2 + \frac{4d(9-d)}{3(4+pv)^2}j_*^2.\end{aligned} \tag{9}$$

For $0 \le p \le N$, the last coefficient is at least $32d/[3(4+N)^2]$. The shifted squares control the other terms with polynomial losses in N. They do not give a lower bound independent of d for the unweighted axial Hessian as $d \downarrow 0$. This distinction matters both for unrestricted scalar inputs and for estimates uniform in N.

Writing $\tau = e^{-3\varepsilon N/2}$, the original three-dimensional source is

$$\Xi = \delta_0\mathcal{T} + \rho + \delta_0\tau\frac{l|a|^2}{\sqrt{D}} - m_0(t)C_N(\rho_0 + v\rho_1), \quad m_0(t) = m(N(t+\varepsilon)). \tag{10}$$

Here $\rho_0 \asymp r^{-4}$, $\rho_1 \asymp r^{-2\gamma}$, $3/4 < \gamma < 1$, and m is the separate penalty cutoff. Along the homotopy, $\mathcal{T}$ uses the actual trace $\mathcal{F}$ even when the scalar prescription changes. Replacing it by a convenient positive quadratic form changes the equation.

3 Full Differential and Exact Ram Reduction

Let (φ, ζ) vary (f, Z), let $h = \nabla\varphi$, and write a dot for this variation. Define $\alpha = \dot{t}$. Implicit differentiation of Z gives

$$\begin{aligned}
\alpha &= \frac{c\zeta - l^2\langle a, h\rangle/D}{c + pv}, & \dot{p} &= p'(t)\alpha, \\
\dot{D} &= 2l^2(\langle a, h\rangle + p|a|^2\alpha), & \dot{d} &= -\dot{D}/D^2, \quad \dot{v} = \dot{D}/D^2, \\
\dot{w} &= \frac{l}{\sqrt{D}}h + \left(p\alpha - \frac{\dot{D}}{2D}\right)w, & \frac{\dot{u}}{u} &= (p+2)\alpha + \frac{\dot{D}}{2D}, \\
\dot{b} &= \frac{\dot{p}(c+pv) - (c+p)(v\dot{p} + p\dot{v})}{(c+pv)^2}, & \dot{A}_\chi &= -\dot{b}\, w \otimes w - b(\dot{w} \otimes w + w \otimes \dot{w}), \\
\dot{H}_f &= \frac{l}{\sqrt{D}}\nabla^2\varphi + \left(p\alpha - \frac{\dot{D}}{2D}\right)H_f, & \dot{\mathcal{F}} &= \dot{A}_\chi : S + A_\chi : \dot{H}_f.
\end{aligned} \tag{11}$$

In particular $\nabla\dot{t} = \nabla\alpha$ differentiates the entire quotient, including its spatially varying coefficients. It is not obtained by retaining only $\nabla\zeta$. The flux derivative is

$$\dot{V}_\lambda = (\dot{u}A_\chi + u\dot{A}_\chi)(c\nabla Z + \lambda K w) + uA_\chi(c\nabla\zeta + \lambda K\dot{w}). \tag{12}$$

For completeness the source derivative of (10) is

$$\begin{aligned}
\dot{\Xi} = \delta_0\dot{\mathcal{T}} + \delta_0\tau\frac{l}{\sqrt{D}}\left[2\langle a, h\rangle + |a|^2\left(p\alpha - \frac{\dot{D}}{2D}\right)\right] \\
- C_N\{m_0'(t)\alpha(\rho_0 + v\rho_1) + m_0(t)\rho_1\dot{v}\}.
\end{aligned} \tag{13}$$

All weights are fixed background functions here. There is no variation of N, R, or the homotopy parameter.

An axis-free formula makes $\dot{\mathcal{T}}$ unambiguous even at $a = 0$. In dimension three put

$$Q = S + p(dt \otimes w + w \otimes dt), \quad P_Q = Q - (\operatorname{tr} Q)g, \quad B(Q, Q) = |Q|^2 - (\operatorname{tr} Q)^2.$$

Expansion of (8) gives

$$\mathcal{T} = \tfrac{1}{2}B(Q, Q) + 2P_Q(w, \nabla t) + 4(p+1)|dt|^2_{A_d} + \mathcal{F}\{\mathcal{F} + bS(w, w) + 2(p+1)w(t)\}. \tag{14}$$

Thus set $U = \mathcal{F} + bS(w, w) + 2(p+1)w(t)$ and differentiate by

$$\begin{aligned}
\dot{Q} &= \dot{H}_f + \dot{p}(dt \otimes w + w \otimes dt) + p(d\alpha \otimes w + w \otimes d\alpha + dt \otimes \dot{w} + \dot{w} \otimes dt), \\
\dot{\mathcal{T}} &= Q : \dot{Q} - (\operatorname{tr} Q)(\operatorname{tr}\dot{Q}) + 2P_{\dot{Q}}(w, \nabla t) + 2P_Q(\dot{w}, \nabla t) + 2P_Q(w, \nabla\alpha) \\
&\quad + 4\dot{p}|dt|^2_{A_d} + 4(p+1)\{2A_d(dt, d\alpha) - 2w(t)\dot{w}(t)\} + \dot{\mathcal{F}}U + \mathcal{F}\dot{U}, \\
\dot{U} &= \dot{\mathcal{F}} + \dot{b}S(w, w) + b\dot{H}_f(w, w) + 2bS(\dot{w}, w) + 2\dot{p}\, w(t) + 2(p+1)\{\dot{w}(t) + w(\alpha)\}.
\end{aligned}$$

The notation $\dot{w}(t)$ means $dt(\dot{w})$, not the variation of $w(t)$; the latter also contains $w(\alpha)$. This gives a full coefficient-aware differential without a singular choice of axis.

3.1 Nonlinear boundary derivative

At an inner face set

$$J_\lambda = 1-(1-\lambda)j(t), \quad B = -H + w_\nu(\mathcal{P}-P_B) - N_B d, \quad C_\lambda = B - (1-\lambda)(A_\chi K w)_\nu.$$

Before the last stage, $q = uJ_\lambda C_\lambda$. Its exact derivative is

$$\begin{aligned} \dot q &= \dot u J_\lambda C_\lambda - u(1-\lambda)j'(t)\alpha C_\lambda + uJ_\lambda \dot C_\lambda, \\ \dot C_\lambda &= \dot w_\nu(\mathcal{P}-P_B) + w_\nu \dot{\mathcal{P}} - N_B \dot d - (1-\lambda)(\dot A_\chi K w + A_\chi K \dot w)_\nu. \end{aligned} \tag{15}$$

The background H, P_B, N_B, ν are not varied. If $h_0 = \tau f$ is the source's scaled height, then at stage III

$$\dot{\mathcal{P}} = \tau\varphi + (1-\zeta_s)\{D_{0,w}\dot w + D_{0,h}\tau\varphi\} + \zeta_s\{\dot A_\chi : K + D_{1,w}\dot w\},$$

where $\zeta_s = s-2$ is held fixed and D_0 is evaluated at $(x, w, h_0+\zeta_s b(x))$. At stage II the corresponding factor is $s-1$, and at stage I $\dot{\mathcal{P}} = \tau\varphi$. At stage IV multiply both $\dot\Xi$ and $\dot q$ by $4-s$. This differentiation retains collar, observation-state, and boundary terms. Consequently

$$D\mathcal{R}_2[\varphi,\zeta][\psi] = \int \dot V_\lambda \cdot \nabla\psi + \int (\dot u \Xi + u\dot\Xi)\psi + \int_{\partial_-\Omega_R} \dot q\psi.$$

Proposition 3.1 (Restricted scalar elimination)**.** *On a fixed smooth truncation take a differentiability neighborhood in Hölder spaces where the scalar Dirichlet realization $A = D_f\mathcal{R}_1$ is a bounded isomorphism and the source's scalar solution map $f = \mathcal{S}(Z)$ is defined. Use homogeneous correction traces after lifting the fixed boundary values. Then*

$$D\mathcal{S} = -A^{-1}D_Z\mathcal{R}_1, \qquad DG = D_Z\mathcal{R}_2 - D_f\mathcal{R}_2A^{-1}D_Z\mathcal{R}_1, \quad G(Z) = \mathcal{R}_2(\mathcal{S}(Z), Z).$$

If $G(Z) = \mathcal{A}(Z)Z - \mathcal{L}(Z)$, where $\mathcal{A}(Z)$ is the frozen mixed weak operator and $\mathcal{K}(Z) = \mathcal{A}(Z)^{-1}\mathcal{L}(Z)$ is differentiable on the same compatible spaces, then at a fixed point

$$DG = \mathcal{A}(Z)(I - D\mathcal{K}(Z)).$$

Proof. Differentiate $\mathcal{R}_1(\mathcal{S}(Z), Z) = 0$ and solve with A^{-1}. Substitute this derivative into the chain rule for G. Finally $G(Z) = \mathcal{A}(Z)(Z - \mathcal{K}(Z))$; its product-rule term $D\mathcal{A}[\zeta](Z-\mathcal{K}(Z))$ vanishes at a fixed point. No positivity of DG follows from positivity of $\mathcal{A}$. □

For example, one may use $f \in C^{3,\beta}$, $Z \in C^{2,\beta}$ locally at smooth solutions, with the corresponding strong residual and trace spaces; the proposition requires their actual bounded mapping and inverse properties. It does not assert that the entire coupled derivative is invertible on the source's degree space $C^{1,\beta}$. Scalar uniqueness is not coupled uniqueness.

3.2 What the Ram representation preserves

The actual private TR8 proof of the weak inverse first specifies the boundary realization, proves coercivity there, and only then permits a Green representation. For M521 the frozen form is $\int cuA_\chi\nabla z\cdot\nabla\psi$ on functions vanishing on the outer boundary; the inner flux remains a load. On a fixed truncation and bounded trial set, ellipticity and Poincaré give its weak inverse. They do not give an N-uniform inverse of DG. If this inverse admits a kernel or distributional kernel $G_Z(x,y)$, Ram localization $y = x - a$ rewrites its action with kernel $G_Z(x, x-a)$ and the transformed domain

and physical volume density. Boundary-load kernels retain their surface measure. This is an exact coordinate identity for that inverse, not a free-space replacement.

The complete TR6 variation proof retains both source-state and observation-state derivatives. Equations (11)–(15) implement that rule for this derivative-dependent system. The actual TR1 stability proof requires an isomorphism on the complete residual space for a root certificate; a finite left inverse cannot silently annihilate boundary or unobserved residuals. The nonlinear-jet proof in the revised TR1 gives the omitted closure term $f^{(k+1)}\partial_{F_k}\Phi$. Small factors bound that term; they do not make it vanish. No polynomial closure of $l(t)$, $D^{-1/2}$, or $m_0(t)$ is assumed here. A complete conforming coordinate change preserves the equations; a finite Ram panel bank is a discretization with unresolved complement and trace errors.

4 Reflected Flux and Natural Growth

The selected existence proof does not differentiate measurable axial ellipticity. It first obtains gradient Hölder control and a Hessian Morrey bound for the graph, then treats Z by a scalar natural-growth argument. The measure carried by reflection is part of that argument.

Proposition 4.1 (Interface identity and exponential product rule)**.** *Let Z be continuous and piecewise C^2 up to the plane $\Sigma = \{x_n = 0\}$ in a ball, with piecewise C^1 coefficients a, b, a symmetric uniformly positive. Write $Q = a\nabla Z + b$, and define the jump using the same positive coordinate direction on both sides, $j = Q_n^+ - Q_n^-$. Then*

$$e := \div Q = e_{\text{bulk}}\, dx + j\mathcal{H}^{n-1}\lfloor\Sigma.$$

For $s \in \{-1, 1\}$, $k > 0$, $V = e^{skZ}$, and $\mathcal{L} = -\div(a\nabla)$, the distribution identity is

$$\mathcal{L}V = \div(skVb) - skVe - k^2V\{a\nabla Z\cdot\nabla Z + b\cdot\nabla Z\}. \tag{16}$$

If the lower half is formed from the upper half by $Z^-(x) = \sigma Z^+(Ex)$, $a^-(x) = Ea^+(Ex)E$, $b^-(x) = \sigma Eb^+(Ex)$, where $E = \operatorname{diag}(1, \dots, 1, -1)$ and $\sigma = \pm 1$, then $j = (1+\sigma)Q_n^+$. The odd case requires zero trace of Z for continuity.

Proof. Integrate $-Q\cdot\nabla\psi$ on the two half balls. The outward normals there are $-e_n$ and $+e_n$, respectively, producing $(Q_n^+ - Q_n^-)\psi$ on the plane. For the nonlinear identity, $\nabla V = skV\nabla Z$, so

$$-\div(a\nabla V) = -skV\div(a\nabla Z) - k^2Va\nabla Z\cdot\nabla Z.$$

Insert $\div(a\nabla Z) = e - \div b$ and expand $skV\div b = \div(skVb) - k^2Vb\cdot\nabla Z$. Continuity makes the trace of V identical on both sides, so multiplication of the plane measure is by that common trace. In particular the interface coefficient on the right is $skV[b_n] - skV[Q_n] = -skV[(a\nabla Z)_n]$, exactly the coefficient on the left. Reflection gives $Q^- = \sigma EQ^+(Ex)$, hence $Q_n^- = -\sigma Q_n^+$. These calculations prove both assertions, including the atom's sign. □

For a curved face the calculation is made in normal coordinates with the volume density incorporated into a, b, Q, e. It is not a Euclidean flux formula with the density dropped. The proposition proves the product rule in a concrete continuous, piecewise smooth class. For merely bounded $W^{1,2}$ fields and measure data, the source explicitly assumes an appropriate representative and weak exponential product rule. We do not remove that hypothesis by formal multiplication.

4.1 Absorbing the quadratic gradient term

Suppose $|Z| \le M$, $\lambda I \le a \le \Lambda I$, $|b| \le B$, and

$$|e| \le C_0(1 + |\nabla Z|^2)\, dx + \mu, \qquad \mu(B_r) \le C_\mu r^{n-2+\beta}, \quad \beta > 0.$$

Under the product rule above choose $k \ge \max(1, 4C_0/\lambda)$. Young's inequality gives

$$k^2 B|\nabla Z| \le \tfrac{1}{2}k^2\lambda|\nabla Z|^2 + \frac{k^2B^2}{2\lambda}.$$

Consequently the total gradient-square coefficient on the right of (16) is at most $kC_0 - k^2\lambda/2 \le -k^2\lambda/4$. Dropping this nonpositive term yields

$$\mathcal{L}(e^{skZ}) \le \div F_s + C(dx + \mu), \qquad F_s = ske^{skZ}b, \quad |F_s| \le ke^{kM}B. \tag{17}$$

This is a proved algebraic absorption, not a regularity theorem without the measure and product-rule assumptions.

Proposition 4.2 (Surface-integrability extension of the potential budget)**.** *Let $n \in \{3, 4\}$ and let Σ be a plane or a uniformly Lipschitz face with $\mathcal{H}^{n-1}(\Sigma \cap B_r(x)) \le C_\Sigma r^{n-1}$. Suppose a local Dirichlet Green kernel satisfies $|G(x, y)| \le C_G|x - y|^{2-n}$. If $j \in L^p(\Sigma)$, $p > n - 1$, and $\|j\|_p \le J$, put $\beta_p = 1 - (n - 1)/p > 0$. Then*

$$\int_{\Sigma\cap B_r(x)} |j|\, d\mathcal{H}^{n-1} \le C_\Sigma^{1-1/p} J r^{n-2+\beta_p}, \tag{18}$$

$$\int_{\Sigma\cap B_R(x)} |G(x, y)j(y)|\, d\mathcal{H}^{n-1}(y) \le \frac{C(n, C_G, C_\Sigma)JR^{\beta_p}}{1 - 2^{-\beta_p}}.$$

Combining this with a bulk measure bound of exponent $n - 2 + 2\alpha$ gives a total exponent $\beta = \min(2\alpha, \beta_p, 1) > 0$ for $r \le 1$.

Proof. Hölder's inequality bounds the first integral by $J[\mathcal{H}^{n-1}(\Sigma\cap B_r(x))]^{1-1/p}$. The exponent equals $(n-1)(1-1/p) = n-2+\beta_p$. Split the second integral into annuli $2^{-m-1}R < |x-y| \le 2^{-m}R$. On the mth annulus the kernel is at most $C_G(2^{-m-1}R)^{2-n}$ and the mass is at most $C_\Sigma^{1-1/p}J(2^{-m}R)^{n-2+\beta_p}$. Their product is bounded by $CJR^{\beta_p}2^{-m\beta_p}$. Sum the convergent geometric series. The same summation applies to the bulk measure. On $r \le 1$ each higher exponent is bounded by the smallest exponent, proving the last statement. □

This is an own restricted extension of the bounded-flux estimate used by the source, with a complete proof. To infer Hölder regularity for the coupled problem with this weaker flux, one must still establish that its actual nonlinear boundary law has this L^p bound, retain the weak product rule, and apply the source's natural-growth lemma. The estimate does not manufacture these hypotheses.

Proposition 4.3 (The critical exponent does not suffice for this budget)**.** *For either $n = 3$ or $n = 4$, there is a nonnegative compactly supported density $j \in L^{n-1}(\mathbb{R}^{n-1})$ whose Newton potential at the origin is infinite.*

Proof. On the plane set $j(y) = |y|^{-1}[\log(1/|y|)]^{-3/4}$ for $0 < |y| < e^{-2}$ and zero otherwise. Writing $m = n - 1$, its L^m integral is a constant times

$$\int_0^{e^{-2}} \frac{dr}{r[\log(1/r)]^{3m/4}} = \int_2^\infty u^{-3m/4}\, du < \infty,$$

because $m = 2$ or 3. Its potential at zero, with kernel $|y|^{2-n}$, is a constant times $\int_2^\infty u^{-3/4}\,du = \infty$. □

This example obstructs a uniform finite Green-potential bound from the critical surface norm alone. It is not a counterexample to the source's natural-growth lemma, which assumes a bounded solution and a positive Morrey excess exponent, and it is not a nonexistence example for M521.

4.2 What the selected natural-growth proof actually does

The source solves zero-boundary correction problems for the two exponential subsolutions. The bounded divergence input contributes $O(R)$, by the relevant weak elliptic estimate; the positive measure contributes $O(R^\beta)$ through its Green bound, and bounded volume input contributes $O(R^2)$. No pointwise gradient bound for a measurable-coefficient Green kernel is presumed here. If $M_R = \operatorname{ess\,sup}_{B_R} Z$ and $m_R = \operatorname{ess\,inf}_{B_R} Z$, corrected versions of $e^{kM_R} - e^{kZ}$ and $e^{-km_R} - e^{-kZ}$ are nonnegative supersolutions. On at least half of the smaller ball, one of the two exponential deficits controls the oscillation. The weak Harnack inequality then gives

$$\operatorname{osc}_{B_{R/4}} Z \le (1-\theta)\operatorname{osc}_{B_R} Z + C(R + R^\beta), \qquad \theta > 0.$$

For any $\gamma < \min(1, \beta, -\log(1-\theta)/\log 4)$, iteration gives $\operatorname{osc}_{B_r} Z \le Cr^\gamma$: divide the recurrence by R^γ and sum a geometric series since $(1-\theta)4^\gamma < 1$. The outcome is a Hölder representative; changing arbitrary point values of the original Borel representative is not ruled out by an a.e. equation. Weak Harnack and the correction estimates are classical analytic inputs to this reconstruction, not newly proved here.

5 Why the Three- and Four-Dimensional Proofs Differ

The axial equation has eigenvalues $1, \dots, 1, \chi$, with $\chi \ge \chi_0 > 0$ only after an appropriate gradient bound. Its coefficient defect satisfies $\|I - A_\chi\|_F = 1 - \chi$. The source uses a Calderón–Zygmund estimate with norm close to one at exponent two, choosing $p_0 > 2$ sufficiently near two. It does not claim a dimension-free Cordes condition for an arbitrary elliptic matrix.

For a scalar graph z the exact cancellation is

$$Q_z = (\nabla^2 z - \Delta_{gz}\, g)\nabla z = A_\chi \nabla(|\nabla z|^2/2) - G\nabla z,$$

$$\operatorname{div}_g Q_z = |\nabla^2 z|^2 - (\Delta_g z)^2 + \operatorname{Ric}(\nabla z, \nabla z).$$

No derivative of χ or G appears in the second identity. The deficit argument forces Hessian energy to zero in a near-maximal-gradient blow-up limit, hence an affine limit. Fixed-axis differentiation then gives the divergence equation $\Delta_\perp v + \partial_e(\chi \partial_e v) = 0$ for $v = \partial_e Y$. Harnack, Caccioppoli, and the Newton-potential estimate transfer its control to the full gradient.

At a constant Dirichlet face the graph is extended oddly in normal coordinates. The extension is C^1 but need not be C^2; no Hessian atom is introduced. The geometric flux is nevertheless not continuous: $Q_z \cdot \nu = -H_B(\nu z)^2$. In dimension four the density-weighted jump in the doubled normal coordinate is $-2\sqrt{\det g}\, H_B(\partial_n z)^2$. It is represented as the divergence of a bounded step field. This is distinct from the even reflection of Z, whose total normal flux produces a surface measure as in Proposition 4.1.

5.1 The borderline dimension-four step

The four-dimensional proof obtains W^{2,p_0} with $p_0 > 2$, but its small correction is initially only small in $W^{2,2}$. It does not use the false embedding $W^{2,2}(\mathbb{R}^4) \subset C^0$. Instead it supplies a common $C^{0,\eta}$ modulus, with $\eta < \min(1, 2 - 4/p_0)$, from the original W^{2,p_0} control and the regular comparison solution. For a correction v with Hölder bound H and $\|v\|_2 \le \epsilon_1$, a point with $|v| = m$ forces $|v| \ge m/2$ on a ball of radius comparable to $(m/H)^{1/\eta}$. Integrating its square gives

$$\epsilon_1^2 \ge c m^{2+4/\eta} H^{-4/\eta}, \qquad m \le C H^{2/(\eta+2)} \epsilon_1^{\eta/(\eta+2)},$$

provided that ball fits in the fixed patch; otherwise the fixed-patch L^2 bound gives the additional linear term $C\epsilon_1$. This elementary estimate explains the extra modulus requirement. The dimension-three argument cannot be transplanted by merely replacing a subscript.

5.2 Closing the coupled bootstrap without circularity

At fixed N, the selected proof proceeds as follows: axial graph estimates give $\nabla f \in C^{0,\alpha}$ and $\int_{B_r} |\nabla^2 f|^2 \le C r^{n-2+2\alpha}$; natural growth then gives $Z \in C^{0,\gamma}$; scalar Schauder estimates give $f \in C^{2,\theta}$; a small-oscillation estimate absorbs the quadratic $|\nabla Z|^2$ term in local $W^{2,p}$ bounds. The normal datum depends only on $Z, f, \nabla f$. Its tangential derivative costs ∇Z and $\nabla^2 f$, not $\nabla^2 Z$ in the boundary data.

The interpolation estimate has the scaling-correct lower term

$$\|\nabla Z\|^2_{L^{2p}(B_h)} \le C \operatorname{osc}_{B_{2h}} Z \, \|\nabla^2 Z\|_{L^p(B_{2h})} + C h^{n/p-2} (\operatorname{osc}_{B_{2h}} Z)^2.$$

It cannot be deleted, as a nonconstant affine function has nonzero gradient but zero Hessian. Choose a uniform small patch size h using the Hölder modulus, then use a bounded-overlap cover and the supremum of patch norms. The absorbing coefficient depends on the overlap count, not on the total number of patches as $R \to \infty$. The supremum is finite for each already smooth finite-truncation solution before absorption. The separate all-order unit-patch hypotheses justify the later Schauder iterations. The maximal chapter's selected regularity proof uses the same local mechanism with the actual decaying K drift; compact support of K is not needed for these local steps. This does not prove its global return-map estimates or its asymptotic flux conclusion.

6 Fixed-Truncation Degree and the Order of Limits

The complete selected three-dimensional degree proof has two logically different estimate regimes. First, for fixed $N \ge 4$, fixed $R < \infty$, and arbitrary trial $Z \in C^{1,\beta}$ with outer value zero, the scalar Dirichlet problem is solved without imposing a floor. Second, a priori estimates and floor exclusion apply to fixed points of the coupled homotopy. Applying the second regime to arbitrary trial functions would leave the compact map undefined on the necessary ambient space.

6.1 The unrestricted scalar solve

For large graph gradient $\sigma = |\nabla f|$, the negative branch of the implicit transformation satisfies

$$e^{8Z} = e^{8t} + e^{(2N+8)t} \sigma^2, \qquad t = -\frac{\log \sigma}{N+4} + O(1).$$

Thus $l \asymp \sigma^{-N/(N+4)}$ and $\chi \asymp \sigma^{-8/(N+4)}$ on bounded trial sets. For $N \ge 4$, this gives a lower bound of the form $c/(1+\sigma)$, not a positive constant independent of unrestricted σ. Height comparison

uses monotonicity of the prescribed trace in $h_0 = \tau f$, with derivative at least one. The logarithmic boundary barrier $\psi(r) = k^{-1}\log(1 + Br)$ satisfies $\psi'' = -k(\psi')^2$. The source chooses k, then the collar width, then B; the order is needed for both steep slopes and sufficient barrier height. Its two-point interior argument uses the sum of the two axial coefficients and does not assume a Lipschitz modulus of χ in the large-gradient regime. Interpolation to $\Delta f = \tau f$ uses the maximum principle for the scalar linearization with a strictly negative zero-order coefficient. It gives a scalar solution map $\mathcal{S}_s$ and uniqueness at fixed input Z. All these constants may depend on N, ε, τ and the fixed truncation. None of this supplies a contraction of the coupled map.

6.2 The four stages and the terminal map

The source first removes the K drift ($0 \le s \le 1$), then inserts the bounded monotone collar penalty D_0 ($1 \le s \le 2$), then deforms the scalar prescription to $h_0 + \operatorname{tr}_{A_\chi} K + D_1$ while its inner Dirichlet data go to zero ($2 \le s \le 3$). The collar directional cutoffs enforce the two possible gradient-direction barriers; the black and white contributions have different signs. Finally it multiplies *both* the scalar-divergence source and the prescribed inner flux by $4 - s$ ($3 \le s \le 4$). At the final scalar stage, cancellation of $\operatorname{tr}_{A_\chi} K$ leaves $A_\chi : H_f = h_0 + D_1$. Since $D_1(x, 0) = 0$, the maximum principle gives $f = 0$ for every trial Z. At $s = 4$ the remaining mixed problem has zero volume source and zero inner flux, so its unique outer-zero solution is zero. Hence $\mathcal{K}_4$ is the zero map on its entire domain, not merely zero at known fixed points.

6.3 The compact map and moving admissible slices

Let $X_\beta = \{Z \in C^{1,\beta}(\overline{\Omega}_R) : Z|_{\partial_+\Omega_R} = 0\}$. For a trial Z, first set $f = \mathcal{S}_s(Z)$ and invert the implicit transformation. Freeze the resulting coefficients and solve

$$\operatorname{div}_g(cuA_\chi \nabla \widetilde{Z} + B_s) = E_s, \qquad (cuA_\chi \nabla \widetilde{Z} + B_s) \cdot \nu = q_s, \qquad \widetilde{Z}|_{\partial_+\Omega_R} = 0.$$

Here B_s is the retained drift, E_s the homotopy-scaled $u\Xi$, and q_s its matching inner flux. The weak load has the signs

$$\int cuA_\chi \nabla \widetilde{Z} \cdot \nabla \psi = -\int B_s \cdot \nabla \psi - \int E_s \psi - \int_{\partial_-\Omega_R} q_s \psi.$$

On bounded trial sets the coefficient is uniformly elliptic with finite constants, and the nonempty outer Dirichlet boundary supplies Poincaré. Regularity improves the output to $C^{2,\beta}$, compactly embedded into X_β. This is the compact return map $\mathcal{K}_s$ of the source.

Its admissible set is the relative open set

$$\mathcal{U} = \{(s, Z) : \min_{\overline{\Omega}_R} t(x, Z, \nabla \mathcal{S}_s(Z)) > -\varepsilon,\ \|Z\|_{C^{1,\beta}} < M_1\}.$$

The floor is evaluated using the scalar solution, not an independently assigned graph. Bounds on admissible coupled solutions exclude fixed points on the norm boundary; the separate floor argument excludes fixed points on the remaining boundary. Both exclusions are indispensable inherited hypotheses for the degree computation.

Proposition 6.1 (Degree on moving slices; reconstruction)**.** *Let $T : [0, 4] \times X \to X$ be continuous and compact on bounded sets, and let $\mathcal{U} \subset [0, 4] \times X$ be bounded and relatively open. Suppose no fixed point lies on its relative boundary. Then $\deg(I - T_s, \mathcal{U}_s, 0)$ is constant in s, with the degree of an empty slice interpreted as zero. If $T_4 = 0$ and $(4, 0) \in \mathcal{U}$, the degree is one and every slice contains a fixed point.*

Proof. The fixed-point set in $\overline{\mathcal{U}}$ is compact: a bounded sequence of fixed points has a convergent subsequence by compactness of T and compactness of $[0,4]$, and continuity identifies its limit. Boundary exclusion places all limits inside $\mathcal{U}$. Near any parameter s_0, finitely many small balls cover the fixed points of the slice, with their closed parameter-cylinder neighborhoods contained in $\mathcal{U}$. A sequence of fixed points outside that cover at parameters tending to s_0 would converge to a fixed point of the central slice outside it, a contradiction. Thus a fixed bounded spatial neighborhood contains all nearby fixed points and has no fixed points on its boundary. Excision followed by ordinary homotopy invariance on this fixed neighborhood proves local constancy. If the central slice contains no fixed point, compactness instead excludes fixed points for nearby parameters and both degrees are zero. Connectedness of the parameter interval gives constancy. At $s = 4$, normalization gives degree one since the only zero of the identity is inside the slice. Nonzero degree gives existence. □

This proof uses classical Leray–Schauder degree properties. It is a reconstruction of the source's argument, not a new degree theorem. It supplies neither uniqueness nor an algorithmic rate.

6.4 Quantitative outer-floor buffer

The source chooses R after N, with $R_{\min}(N)$ at least N, the geometric and a priori thresholds, and

$$\left[\frac{2C_{\text{out}}\ell}{\tau\sqrt{e^{8\varepsilon}-1}}\right]^{1/(1+\gamma)}, \qquad \ell = e^{-\varepsilon N}. \tag{19}$$

Indeed the outer gradient estimate gives $\sigma \le C_{\text{out}}\tau^{-1}R^{-1-\gamma}$. At $t = -\varepsilon$, $Z = 0$ would require $\ell^2\sigma^2 = e^{8\varepsilon} - 1$. The threshold (19) gives the strictly smaller bound $\ell^2\sigma^2 \le (e^{8\varepsilon} - 1)/4$ and therefore excludes outer contact with the floor. This is a check of the displayed buffer, not a derivation of the inherited outer gradient estimate or the other thresholds. The extra requirement $R_{\min}(N) \to \infty$ is used by the source's contradiction argument. Exhaustion then takes $R \to \infty$ at *fixed* N, using estimates uniform in R. Its limit may satisfy only $t \ge -\varepsilon$. End normalization, scalar integrability, the energy flux, and a subsequent $N \to \infty$ argument are separate steps. A compact-local limit alone does not identify ADM mass.

7 Maximality, End Forcing, and a Restricted Extension

The complete transition proof read here isolates why maximality is analytically useful when K is not compactly supported. On the exterior where the original scalar prescription is $\mathcal{F} = \eta_N f$, its exact expansion is

$$A_\chi : \nabla^2 f + B_f \cdot \nabla f - c_f f = -\frac{\sqrt{D}}{l}\operatorname{tr}_g K, \quad B_f = -\frac{bl}{\sqrt{D}}K(\nabla f, \cdot), \quad c_f = \eta_N\frac{\sqrt{D}}{l}. \tag{20}$$

For maximal data the right side vanishes. On the source's bounded fixed-N class, c_f has a positive lower bound and A_χ, B_f are bounded. The comparison barrier e^{-ar} is a supersolution for sufficiently small $a > 0$, with all constants allowed to depend on N. This gives exponential graph decay uniformly in the exhaustion radius. Higher derivative decay uses the separately established uniform local regularity, not the zero-order comparison alone.

Proposition 7.1 (Conditional nonmaximal polynomial graph tail)**.** *On annuli $r_0 < r < R$ in an asymptotically Euclidean end, suppose a C^2 solution of* (20) *has $f = 0$ at $r = R$ and $|f| \le M$*

at $r = r_0$. Suppose, uniformly in R, that A_χ is positive definite with bounded norm, $|B_f| \le B_0$, $c_f \ge c_0 > 0$, $\sqrt{D}/l \le C_0$, and $|\operatorname{tr}_g K| \le Tr^{-\kappa}$ for some $\kappa > 0$. Suppose the end has $|\nabla r| \le C$, $|\nabla^2 r| \le C/r$. After increasing r_0 by a fixed amount if necessary, there is $A < \infty$, independent of R, such that

$$|f| \le Ar^{-\kappa}.$$

The constants may depend on N through the displayed coefficient bounds. The coefficients are evaluated along the given solution; existence and admissibility are not conclusions.

Proof. Let $\mathcal{P} = A_\chi : \nabla^2 + B_f \cdot \nabla - c_f$. The radial derivative formulas and the assumed end bounds give

$$|A_\chi : \nabla^2 r^{-\kappa}| \le C_1 r^{-\kappa-2}, \qquad |B_f \cdot \nabla r^{-\kappa}| \le C_2 r^{-\kappa-1}.$$

Choose r_0 sufficiently large that $C_1 r^{-2} + C_2 r^{-1} \le c_0/2$. Then $\mathcal{P}(r^{-\kappa}) \le -(c_0/2)r^{-\kappa}$. Take $A \ge \max(2C_0T/c_0, Mr_0^{\kappa})$ and $w = Ar^{-\kappa}$. If the right side of (20) is h, then $|h| \le C_0 Tr^{-\kappa}$ and $\mathcal{P}w \le -|h|$. Hence $\mathcal{P}(f - w) \ge 0$ and $\mathcal{P}(-f - w) \ge 0$. Both functions are nonpositive on the two boundary spheres. At a positive interior maximum their gradients vanish, their Hessians are negative semidefinite, and the strictly negative zero-order term forces $\mathcal{P} < 0$, a contradiction. Thus $-w \le f \le w$. Each finite annulus suffices for this maximum principle; no limiting boundary value at infinity has been assumed. □

If r_0 is enlarged, a uniform bound on the new inner sphere is needed; the source's bounded fixed-N solution class supplies it. Without such a bound the proposition must be applied with its stated inner datum. The proof does not promote polynomial decay of f to decay of all derivatives unless the appropriately scaled regularity bounds on coefficients and forcing are also supplied. The obstruction to keeping the original exponential conclusion is genuine: for a model $(\Delta - c_0)f = -r^{-\kappa}$, exponential decay of f and its first two derivatives would make the left side exponential, contradicting the prescribed polynomial forcing. Thus relaxing maximality changes the tail mechanism, even before the global geometric estimates are considered.

7.1 The tensor remains in the zero-graph equation

In the four-dimensional maximal limit $f = 0$ one has $w = 0$, $d = \chi = 1$, $t = Z$, but K is not zero. The transition proof gives

$$\mathcal{T} = \tfrac{1}{2}|K|^2 + 3(p+1)|dZ|^2.$$

Up to exponentially decaying graph errors, the scalar equation is

$$3\Delta Z + 3[p + 2 - \delta_0(p+1)]|dZ|^2 = \tfrac{\delta_0}{2}|K|^2 + \rho - m_0(Z)C_N\rho_0 + E_N.$$

Define $\Phi(0) = 0$ and $\Phi'(z) = \exp \int_0^z [p(s) + 2 - \delta_0(p(s)+1)]\, ds$. Then $\Delta\Phi(Z)$ is $\Phi'(Z)/3$ times the right side, exactly removing the displayed gradient square. With $K = O(r^{-1-q})$ and $\delta < q - 1$, the tensor square decays faster than $r^{-4-\delta}$. The positive barrier $r^{-2} - r^{-2-\alpha}$, $\alpha = \delta/2$, has leading negative Laplacian $\alpha(2+\alpha)r^{-4-\alpha}$ in dimension four. The source uses this on each truncation before passing to infinity, obtains $Z = O(r^{-2})$, and then derives the first and second derivative bounds on rescaled annuli.

The tail weights are dimension-specific. Choosing $1 < \gamma < \min(2, q - \delta/2)$ gives

$$\omega r^{-2-q} \le \tfrac{1}{2}\omega^2 r^{-2\gamma} + \tfrac{1}{2} r^{-4-2q+2\gamma} \le C(v\rho_1 + \rho_0),$$

where $\rho_0 \asymp r^{-4-\delta}$ and $\rho_1 \asymp r^{-2\gamma}$. Also ρ_1^2 is integrable in four dimensions since $4\gamma > 4$. The three-dimensional interval $3/4 < \gamma < 1$ does not satisfy this last condition. Reusing it in the retained-tensor four-dimensional branch would break the tail budget.

8 Primary Literature and the Scope of Comparison

Public primary records were checked on October 8, 2026 using public titles only. This is a targeted comparison, not a systematic novelty search. Han and Khuri's *Existence and Blow-Up Behavior for Solutions of the Generalized Jang Equation* studies existence, regularity, and blow-up, including nonuniqueness of blow-up solutions [2]. That work is relevant to why a scalar graph construction has nontrivial boundary and branch issues, but it does not establish the compact-map or admissibility estimates of this particular deformation.

Jaracz's *Nonexistence of Solutions to the Coupled Generalized Jang Equation/Zero Divergence System* proves a spherical counterexample for that specified coupling and its required asymptotics [3]. M521 instead prescribes a nonzero divergence source, collar modifications, and a floor penalty. The counterexample is therefore a warning about transferring solvability between coupled systems, not a disproof of M521. Conversely, avoiding the literal zero-divergence equation is not itself an existence proof.

The April 2026 version of Williams's *Numerical investigation of the generalized Jang equation coupled to conformal flow of metrics* reports no analogous finite-radius breakdown in its spherical, time-symmetric numerical setting [4]. This is numerical evidence for a different coupling, not a continuum certificate for the present full system or a general Penrose proof.

Bray and Lee's *On the Riemannian Penrose inequality in dimensions less than 8* gives the relevant Riemannian comparison setting [5]. Its stated numerical inequality allows an outer-minimizing horizon with more than one component. The spin assumption in its equality clause is a separate issue from the numerical inequality. Applying a Riemannian theorem requires producing the complete comparison metric, appropriate enclosure, scalar sign, asymptotics, and mass/area transfer first. None of our local differential identities supplies those geometric hypotheses. Riemannian equality describes the comparison geometry, not automatically the original initial data.

9 Benefits, Costs, and Further Work

9.1 Benefits of the exact formulation

The explicit differential identifies which implementation terms must survive: implicit t variation, $p'(t)$, axial coefficient variation, the actual trace in $\mathcal{T}$, the penalty cutoff derivative, and the nonlinear inner-flux derivative. It makes a derivative test diagnostic: an apparent Newton failure can be separated from a missing collar or boundary term. Scalar elimination reduces the number of unknown fields while Proposition 3.1 retains the full feedback through $\mathcal{S}_s$. This is an organizational advantage, not a claim of faster convergence or a measured speedup.

The interface identity records the sign and density of the doubled flux. Proposition 4.2 supplies a concrete weaker sufficient flux condition with an explicit exponent and a constant that displays deterioration near criticality. Proposition 4.3 prevents replacing this condition by an unjustified endpoint assertion. The tail proposition identifies the exact forcing introduced by loss of maximality, instead of concealing it in a generic small perturbation. Each benefit is tied to the actual source operator and its boundary realization.

9.2 Costs and limitations

The constants in the unrestricted scalar solve, Hölder bootstraps, elliptic inverses, and end comparison may depend badly on fixed N, the small height coefficient, and the admissibility margin. The square completion controls a weighted axial direction; it does not uniformly control an unweighted Hessian

as the graph becomes steep. The factor $(1 - 2^{-\beta_p})^{-1}$ diverges as $p \downarrow n - 1$. Such deterioration matters to any numerical tolerance budget.

Boundary preparation, strict trappedness, all-cut area protection, and formation of a minimal enclosure are geometric requirements beyond a local residual. Boundary components and inward-normal conventions cannot be merged into a single arbitrary Robin condition. The degree proof can produce multiple solutions and gives no canonical branch or rate. Uniform-in-R estimates at fixed N are weaker than joint control of N and R. The local maximal argument does not remove the topology, energy, outermost, or all-cut hypotheses of the application. A finite test bank neither proves continuum invertibility nor checks every cut.

9.3 Concrete extensions and their obstacles

First, one could seek a natural-growth theorem for rougher inner flux using the surface L^p budget proved here. The next necessary step is to establish trace integrability of the *actual* state-dependent normal law uniformly along the homotopy. Merely assigning $j \in L^p$ after reflection does not close that feedback. At the endpoint, a stronger logarithmic or potential condition might replace a positive power, but Proposition 4.3 shows why the plain critical norm is insufficient. A Dini-type oscillation budget, rather than the present geometric-series proof, would then be needed.

Second, the polynomial-tail extension suggests a controlled nonmaximal branch with decaying $\operatorname{tr} K$. One must propagate derivative decay, rework the scalar source and penalty weights, retain a strict dominant-energy margin, and check that polynomial graph contributions are integrable and have acceptable mass flux. Proposition 7.1 proves none of those steps. In particular a positive zero-order coefficient depending on N does not yield an N-uniform tail estimate.

Third, an implementable local certificate could use the exact Schur complement together with a verified inverse on the unretained modes and a full trace residual. This requires real continuum inverse bounds, quadrature enclosures, geometry error bounds, and a ball lying strictly inside the admissible floor. Singular values of a finite matrix alone do not meet that requirement. The inherited degree proof may be more robust for existence when such inverse information is unavailable; it does not then become a contraction theorem.

Fourth, one could make the fixed-N regularity constants quantitative. The axial ellipticity, Calderón–Zygmund exponent gap, Harnack constants, reflection geometry, and bounded-overlap absorption would all need explicit propagation. The difficult step is preventing the resulting dependence from growing so fast that it defeats the exponentially small geometric error when $N \to \infty$. This report has not carried out that balance.

10 Contribution and Evidence Boundaries

The realized contribution is C2/5. The principal own restricted proofs are the interface/exponential identity, the L^p surface-flux budget and its critical obstruction, the exact scalar-elimination relation, and the conditional nonmaximal polynomial tail. The local coefficient calculus and the explicit floor-buffer calculation support those results. They do not amount to a global geometric extension of the source theorem. The square completion, classical degree properties, regularity machinery, and maximal transition mechanism are inherited or classical and are not relabeled as new results.

The application is L4 because it uses the PDE-first boundary realization and complete nonlinear derivative from the actual private Ram modules to expose source-specific conditions. No global injectivity, global uniqueness, universal finite-check guarantee, novelty priority, or numerical speedup is claimed. No computer-assisted solution of the actual coupled geometric PDE was performed. The separately recorded checks test algebra, derivative implementation, reflected

signs, exponent thresholds, and model barrier inequalities. Their outcomes and exact tested scope are in `M521-check-results.json`; the continuum claims remain the displayed proofs with their hypotheses.

10.1 Literal read scope

The scientific reads of the pinned build were bounded as follows. Line numbers are inclusive and refer to the bytes hashed in the read-scope receipt, not a claim to have read every file in the composite.

File relative to the build directory	Lines read
`introduction.tex`	1–289
`neutral/00-setting.tex`	1–201
`neutral/02-system.tex`	1–751
`neutral/degree/degree.tex`	1–1175
`transition.tex`	1–254
`four/four-opening.tex`	1–224
`four/four-system.tex`	1–180
`four/four-regularity.tex`	1–857
`maximal/maximal-method.tex`	1–199; 2080–2375

This is 4,426 distinct source lines across nine files. The partial four-dimensional system read checks its variable and scalar-identity normalization, not the complete proof of that identity. The other eleven TeX files and bibliography were inventoried but not scientifically read in full for this report. In particular the complete domain preparation, admissible estimates, exterior flux, four-dimensional existence and geometry, and remaining maximal construction are not certified here. Some are described through the statements and dependency explanations in the files actually read.

The routing maps were inspected for the relevant foundation entries; map descriptions were not used in place of proofs. Actual private proof intervals were TR8 lines 73–216 and 326–424, TR6 lines 317–393, TR1 `stability.tex` lines 67–216, and revised TR1 `sec-03-jets.tex` lines 1211–1305. The first TR8 interval includes the complete coercive and boundary-lift Green-equivalence arguments; the other intervals include the nonlinear variation, root-certificate, coordinate-transfer, and nonlinear-jet proofs used here. The revised jet discussion's phrase about small omitted factors is read as an error bound, not exact vanishing. No private content was submitted in a public search.

10.2 Identity, checks, and unresolved dependencies

The source receipt distinguishes transcript-observed hashes before the scientific reads, final local hashes, and comparisons with actual Git blobs from the pinned and dated-cache commits. Equal observed endpoints do not prove that no transient edit occurred during reading. A dated cache is not a fresh live upstream check. Private proof hashes identify the actual local material consulted; they do not make private mathematical claims independently verified by provenance. The release manifest freezes the scientific source, exported PDF, checks and scripts, read-scope and source-identity receipts, assessment, build evidence, and referenced input identities. The native compiler dispatch and physical export are separate events. Exact outcome receipts identify any native limitation. No successful render is interpreted as a proof check. The unresolved global dependencies remain unresolved even when every release hash matches.

References

[1] OpenAI mathematics source, *Boundary graph deformations for the spacetime Penrose inequality*, September 27, 2026, locally pinned build; commit and literal read scope specified in Section 1 and the evidence receipt.

[2] Q. Han and M. Khuri, *Existence and Blow-Up Behavior for Solutions of the Generalized Jang Equation*, Comm. Partial Differential Equations 38 (2013), 2199–2237. `https://arxiv.org/abs/1206.0079` (v2, January 16, 2014).

[3] J. S. Jaracz, *Nonexistence of Solutions to the Coupled Generalized Jang Equation/Zero Divergence System*, Classical and Quantum Gravity (2023). `https://arxiv.org/abs/2304.09332`.

[4] H. Williams, *Numerical investigation of the generalized Jang equation coupled to conformal flow of metrics*, `https://arxiv.org/abs/2601.15359` (v2, April 20, 2026).

[5] H. L. Bray and D. A. Lee, *On the Riemannian Penrose inequality in dimensions less than 8*, Duke Math. J. 148 (2009), 81–106. `https://arxiv.org/abs/0705.1128`.

[6] M. Subbarao, private Ram corpus: TR8 PDE-first weak realizations and nonlinear derivatives; TR6 full Urysohn variation; TR1 stability; revised TR1 nonlinear jets. Exact local paths, intervals, and hashes are recorded in the confidential read-scope receipt. These are private supporting sources, not public literature.